

The implication of irrigation in climate change impact assessment: a European-wide study

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Abstract

This study evaluates the impacts of projected climate change on irrigation requirements and yields of six crops (winter wheat, winter barley, rapeseed, grain maize, potato, and sugar beet) in Europe. Furthermore, the uncertainty deriving from consideration of irrigation, CO₂ effects on crop growth and transpiration, and different climate change scenarios in climate change impact assessments is quantified. Net irrigation requirement (NIR) and yields of the six crops were simulated for a baseline (1982–2006) and three SRES scenarios (B1, B2 and A1B, 2040–2064) under rainfed and irrigated conditions, using a process-based crop model, SIMPLACE <LINTUL5, DRUNIR, HEAT>. We found that projected climate change decreased NIR of the three winter crops in northern Europe (up to 81 mm), but increased NIR of all the six crops in the Mediterranean regions (up to 182 mm yr⁻¹). Climate change increased yields of the three winter crops and sugar beet in middle and northern regions (up to 36%), but decreased their yields in Mediterranean countries (up to 81%). Consideration of CO₂ effects can alter the direction of change in NIR for irrigated crops in the south and of yields for C3 crops in central and northern Europe. Constraining the model to rainfed conditions for spring crops led to a negative bias in simulating climate change impacts on yields (up to 44%), which was proportional to the irrigation ratio of the simulation unit. Impacts on NIR and yields were generally consistent across the three SRES scenarios for the majority of regions in Europe. We conclude that due to the magnitude of irrigation and CO₂ effects, they should both be considered in the simulation of climate change impacts on crop production and water availability, particularly for crops and regions with a high proportion of irrigated crop area.

Keywords: climate change, CO₂ effects, crop model, irrigation, LINTUL, SIMPLACE, water availability, yield change

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Introduction

In Europe, the prevalence of irrigation varies from north to south (Boreal to Mediterranean) and from west to east (Atlantic to Continental). Agriculture constitutes 64–80% of total water abstractions in Mediterranean countries, while it consumes less than 1% in Western and northern European countries (Rey *et al.*, 2011). In the arid and semi-arid Mediterranean regions, irrigation is used to avoid drought stress throughout the entire growing season (Rodríguez Díaz *et al.*, 2007), while in the humid northwestern regions, irrigation is used to supplement precipitation during sensitive crop growth stages (Katerji *et al.*, 2008). While the expansion of irrigation has played an important role in promoting food production and closing yield gaps in the past decades in Europe (Baldock & Athēnōn, 2000; Lobell *et al.*, 2008), further expansion may be limited by a lack of water resources in many currently irrigated croplands

(Godfray *et al.*, 2010), presenting a significant challenge to meeting increased demand for food and feed expected of an increasingly affluent global population (Carruthers *et al.*, 1997; Hanjra & Qureshi, 2010).

Climate change, characterized by increased atmospheric CO₂ concentration, higher temperatures and variability, more frequent extreme events (e.g. droughts and heat waves) and shifting precipitation patterns, may further alter current water availability for food production (IPCC, 2007; Schmidhuber & Tubiello, 2007; Schewe *et al.*, 2014). Water availability in croplands depends not only on climate and soil conditions but also on crop water uptake and management practices (Tolk *et al.*, 1999; Rodríguez Díaz *et al.*, 2007; Katerji *et al.*, 2008; Rey *et al.*, 2011; Mahajan *et al.*, 2012). Under climate change, alternative crops and/or management practices (e.g. apply irrigation) are likely to be needed to sustain crop production and mitigate environmental impacts. Determination of appropriate crop management adaptations requires consideration of the various interactions between crop growth, environmental conditions, and management practices such as irrigation

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(Müller *et al.*, 2014). Empirical or statistical models may fail to reflect such interactions, particularly for nonlinear processes, while process-based crop models are developed to simulate interactions between biophysical processes and management practices (Williams *et al.*, 1989; Keating *et al.*, 2003; Van Oijen & Lefelaar, 2008; Zhao *et al.*, 2013). The mechanisms represented in process-based crop models allow for studying the dynamics of agricultural systems under changing environmental conditions (Wriedt *et al.*, 2009; Flörke *et al.*, 2010), making them useful tools for assessing the impacts of climate change on crop yields (Rosenzweig & Parry, 1994; Rosenzweig *et al.*, 2002; Ewert *et al.*, 2015) and crop water availability (Zhang & Cai, 2013; Elliott *et al.*, 2014), as well as assessing adaptation strategies to climate change (Fischer *et al.*, 2007). For example, Wriedt *et al.* (2009) estimated current irrigation water requirements for Europe using the EPIC crop model for a range of crops. Trnka *et al.* (2011) found that irrigation requirements will increase under climate change in the Central Mediterranean and the Iberian Peninsula for both winter and spring crops. Elliott *et al.* (2014) found that the CO₂ effects on crop growth and transpiration were a major source of uncertainty for assessment of climate change impacts on water use and crop yields. To our knowledge, a study on the impacts of climate change on irrigation requirements for major crops in Europe considering the uncertainty from CO₂ effects has not yet been conducted.

Despite the importance and prevalence of irrigation in many European regions, a number of previous studies have considered only rainfed production in their modeling assessments of climate change impacts on crop yields (Van Der Velde *et al.*, 2009; Moriondo *et al.*, 2011; Angulo *et al.*, 2013). The results from these studies may be biased for highly irrigated crops and regions. For example, Bondeau *et al.* (2007) have suggested that using geo-referenced irrigation information could improve the simulation of global net primary productivity (NPP) with the vegetation model of LPJ. Alexandrov *et al.* (2002) report climate change led to greater yield increases for irrigated soybean than for rainfed ones in northeastern Austria using the DSSAT crop model. Therefore, to improve the simulation of future climate change impact on cropping systems, it is important to quantify the error made in neglecting irrigation and considering only rainfed production (and their associated levels of water stress and its interactions with other changing climatic variables) in climate change impact assessments (Olesen & Bindi, 2002).

Accordingly, the first objective of this study was to quantify the impact of projected climate change on irrigation requirements and yields for six crops (winter wheat, winter barley, rapeseed, grain maize, potato,

and sugar beet) across Europe. The second objective was to quantify the importance of considering irrigation and CO₂ effects on crop growth and transpiration in simulating projected climate change impacts on crops' water availability and yields with a process-based crop model.

Materials and methods

Study area, simulation unit, soil, and climate data

The study was conducted for Europe (EU25), and input data were combined from three aggregation levels: environmental zone (ENZ) (Fig. 1a) (Metzger *et al.*, 2005), NUTSII zone (Fig. 1b), and simulation unit (SU) (Fig. 1c). Each ENZ has similar environmental conditions, such as topography and climate, and in this study, twelve ENZs were used to aggregate phenology observations. The NUTSII zones are subnational administrative regions of the EU25 for which agricultural census data (e.g. fertilization rates, irrigated/cropping area, and crop yields) are reported. The SU for which simulation was performed was created as the spatial intersection of ENZs and NUTSII regions.

One-layer soil physical properties for each SU were derived from the SEAMLESS database (Van Ittersum *et al.*, 2008; Janssen *et al.*, 2009) and used in the SIMPLACE <LINTUL5, DRUNIR, HEAT> water balance model. The six physical properties include soil depth, as well as the soil water content at air-dry, wilting point, field capacity, saturation, and water logging and are reported in Fig. S1.

Daily weather data (maximum and minimum temperature, precipitation, global solar radiation, and wind speed) for each SU for the baseline period (1982–2006) were extracted from the SEAMLESS database (Van Der Goot & Orlandi, 2003; Van Ittersum *et al.*, 2008; Janssen *et al.*, 2009). Future climate data were generated by downscaling projected climate data from global climate models (GCMs) for three SRES scenarios, B1 (MIROC3.2), B2 (15-GCM ensemble mean), and A1B, 2040–2064 (15-GCM ensemble mean) (Wolf *et al.*, 2012; Angulo *et al.*, 2013). The CO₂ concentration for the baseline, B1, B2, and A1B scenario periods was set at 369, 487, 478, and 532 ppm, respectively. The spatial variability in the differences between the three SRES scenarios and the baseline for mean temperature, solar radiation, and precipitation are reported in Fig. 2. For simplicity, we present results in maps only for simulations in the A1B scenario and the maps for other scenarios are presented in the Supplementary Materials.

Model description

The irrigation requirement and yields for six crops (winter wheat, winter barley, rapeseed, grain maize, potato and sugar beet) were simulated by SIMPLACE <LINTUL5, DRUNIR, HEAT>. The first three crops were autumn-sown winter crops (hereafter referred to as winter crops), and the last three were spring-sown spring crops. SIMPLACE (Scientific Impact assessment and Modeling PLatform for Advanced Crop and Ecosystem management) is developed to enable the dynamic

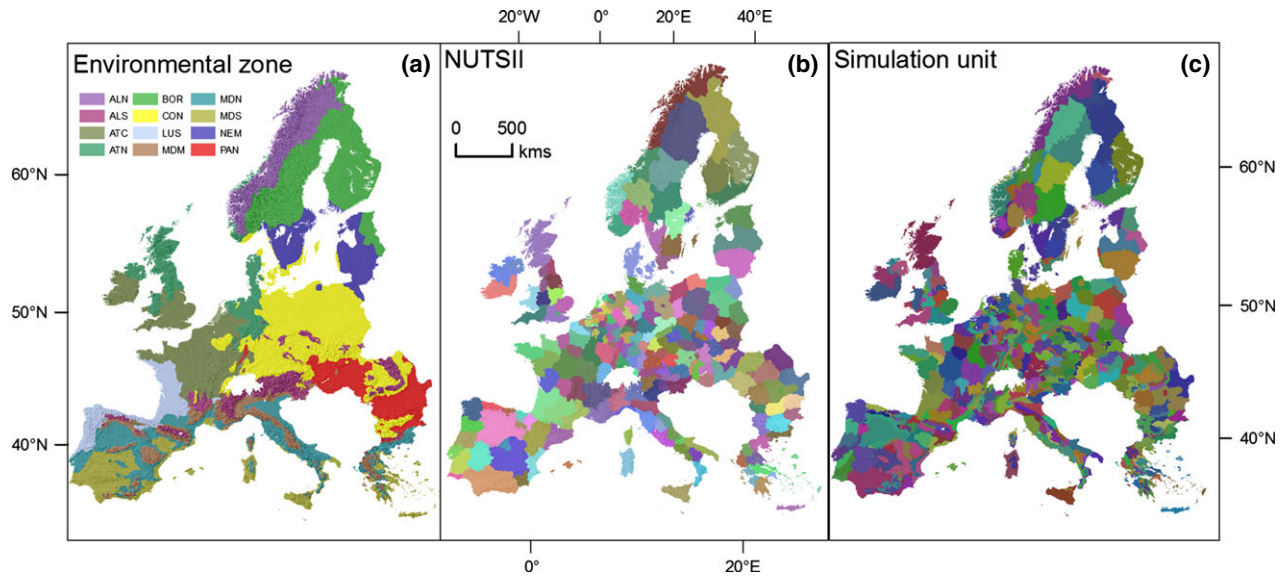


Fig. 1 Three aggregation levels of the study area, European environmental stratification, also known as environmental zone (a, $n = 12$), nomenclature of territorial units for statistics II (b, $n = 240$), and simulation unit (c, $n = 533$). The environmental zone was extracted and modified from Metzger *et al.* (2005) including ALN (Alpine North), ALS (Alpine South), ATC (Atlantic Central), ATN (Atlantic North), BOR (Boreal), CON (Continental), LUS (Lusitanian), MDM (Mediterranean Mountains), MDN (Mediterranean North), MDS (Mediterranean South), NEM (Nemoral), and PAN (Pannonian–Pontic).

linking of models that represent different aspects and processes in agricultural systems for a variety of scientific purposes (Gaiser *et al.*, 2013). Phenology, dry matter production and partitioning, and nitrogen balance modules from LINTUL5 were linked with the soil water balance module (DRUNIR) from LINTUL2 and a newly implemented heat stress module (Eyshi Rezaei *et al.*, 2015), collectively referred to SIMPLACE < LINTUL5, DRUNIR, HEAT >. The model has been widely validated to simulate crop growth and yields in Europe and other regions across the world for a variety of crops (Shibu *et al.*, 2010; Angulo *et al.*, 2013; Asseng *et al.*, 2013; Gaiser *et al.*, 2013; Zhao *et al.*, 2015).

SIMPLACE < LINTUL5, DRUNIR, HEAT > simulates daily dry matter production as the product of intercepted solar radiation and radiation use efficiency (RUE) (Gaiser *et al.*, 2013). RUE varies with the daily mean and minimum temperature, CO₂ concentration, and crop development stage, using piecewise interpolation functions (Fig. S2). Radiation interception is a function of incident photosynthetically active radiation and leaf area index (LAI). Very early LAI development is exponential and limited only by temperature. Later, the daily increase in LAI is constrained by the total biomass partitioned to leaves, specific leaf area (SLA), maximum daily LAI increase, and death rate. The partitioning fraction, SLA, and death rate vary with the crop development stage with parameters from Boon-Prins *et al.* (1993). Crop phenological development is simulated by accounting for thermal time requirement and accumulation, vernalization, and day-length effects, as appropriate (Asseng *et al.*, 2013).

A one-layer tipping bucket approach is used to simulate the soil water balance across available soil water, rainfall, irrigation, drainage, runoff, and evapotranspiration (Shibu *et al.*,

2010). Crop available water is calculated as the product of rooting depth and extractable soil water. The increase in rooting depth is assumed constant until reaching one of either the crop-specific maximum root or soil depth. Water stress occurs when the available soil water is either below a crop-specific critical point or above field capacity (water logging). Water stress reduces dry matter production, the leaf expansion rate, and actual evapotranspiration rate, and affects the root–shoot partitioning ratio. Actual transpiration is calculated as the product of the water stress reduction factor and potential transpiration (Penman, 1963). CO₂ effect on transpiration for different crops was derived from literature on FACE experiments (Kimball *et al.*, 1995; Hunsaker *et al.*, 1996, 2000; Qaderi & Reid, 2005; Kim *et al.*, 2006; Leakey *et al.*, 2009; Burkart *et al.*, 2011; O’leary *et al.*, 2015) and is reported in Tables S2 and S3. In this study, the soil water and nitrogen (N) contents were reinitialized at sowing; thus, no carry over effects among growing seasons were simulated. The crop growth-related parameters, except the thermal requirements for the phenology simulations, for each crop, were obtained from Boon-Prins *et al.* (1993).

Phenology, management, and yield data

Historical sowing, flowering, and harvest dates were extracted from the JRC/MARS phenology database for Europe (Willekens *et al.*, 1998; Angulo *et al.*, 2013). The data are geo-referenced, but are not available for all years or for all NUTSII zones where historical yields and fertilization rates are available. To fill these gaps, each record was treated as an observation for its respective ENZ. The median of observations in each ENZ was calculated and used for calculation of thermal

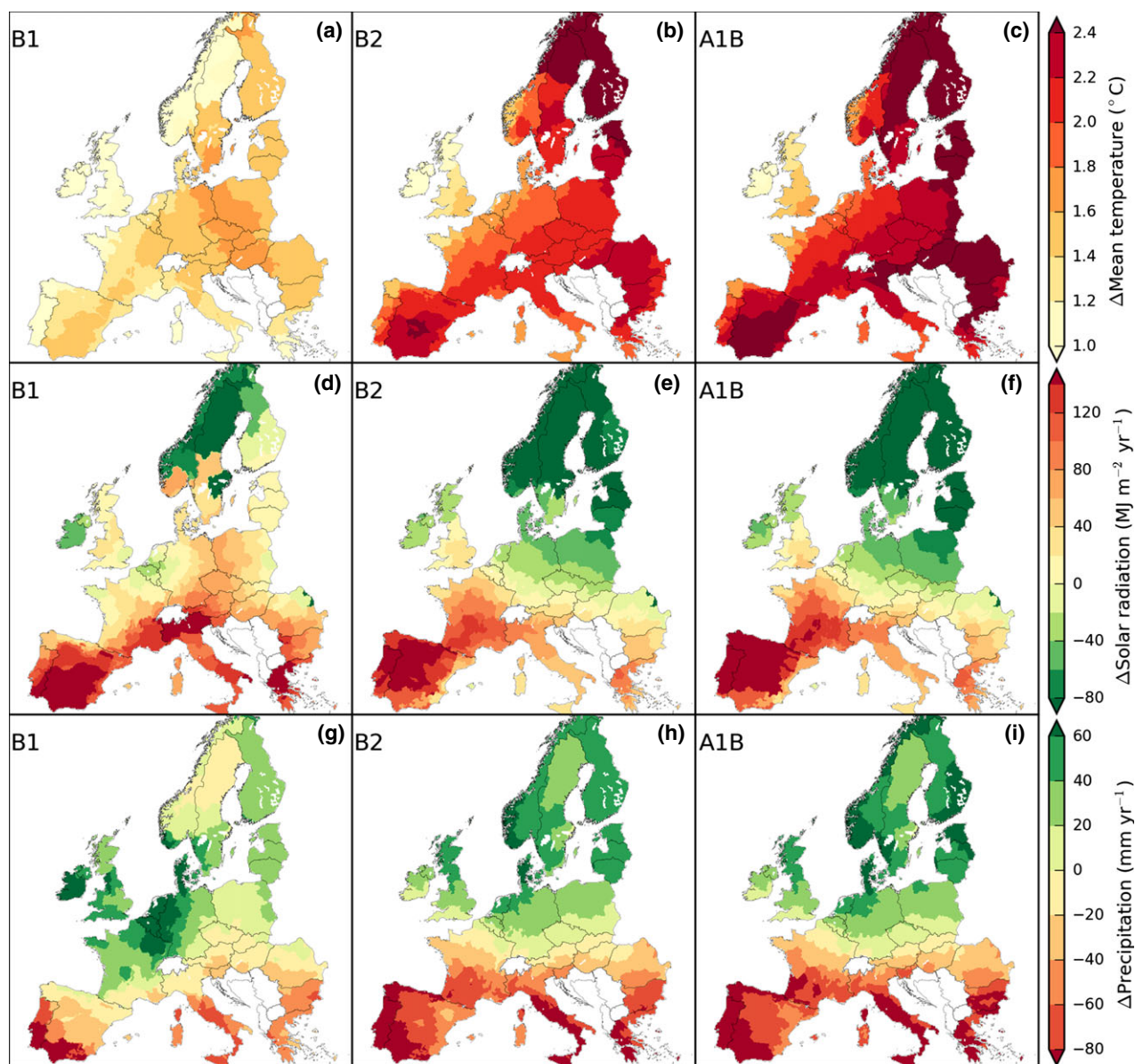


Fig. 2 Changes of weather conditions (mean temperature, global solar radiation, and precipitation) in three SRES climate change scenarios (i.e. B1, B2, A1B). Data for mean temperature (a, b, and c) refer to daily mean, while data for solar radiation (d, e and f) and precipitation (g, h and i) refer to annual sums.

requirements (Fig. S3). When observations for a particular crop and ENZ were missing, the across ENZ averages for each crop were used to generate the missing values by preserving the relationship between the average difference between crops in the ENZ in question. For example, when the sowing date for winter wheat was available in the ENZ of BOR, but was not available for winter barley, the sowing dates of winter wheat and the average difference between these two crops across all ENZ was used to derive the sowing date of winter barley. With the observed dates of sowing, flowering and maturity, and crop base temperatures (Table S1), we computed the thermal requirements ($^{\circ}\text{C d}$) from emergence to flowering and from flowering to crop maturity (Fig. S3).

According to Connor *et al.* (2011), the three winter crops required vernalization and showed obligate long-day photoperiodic response. Temperate grain maize, potato, and sugar beet were assumed to be day neutral. The detailed phenology parameters are reported in Table S1. The sowing dates and the required thermal times in projected climate change scenarios were kept the same as the baseline.

Nitrogen application rates for the six crops at NUTSII level for the period of 1983–2006 were originally from FAO (2009) and postprocessed for the CAPRI database (Britz *et al.*, 2007) (Fig. S4). Missing values for a specific year were filled with the historical mean for the same NUTSII zone (Webber *et al.*, in review). For a crop in a NUTSII zone with no data for the

entire baseline period, simulations were not conducted. As nitrogen application rates were not separately reported for irrigated and rainfed production systems, irrigated crops were allocated twice as much as nitrogen as the rainfed ones, with the total amount of nitrogen applied to irrigated and rainfed crops summing to the reported value for the NUTSII zone. A detailed description of nitrogen simulation setup can be found in Webber *et al.* (in review).

Irrigation requirement and climate change impacts

Following the approach of Allen *et al.* (1998), Smith (1992), and Döll (2002), a depth of net irrigation requirement (NIR, mm) for each simulation unit (SU) was calculated on a daily time step during the growing season as the difference between potential evapotranspiration (pET, mm) under fully irrigated conditions and effective precipitation (PE, mm).

$$\text{NIR} = \text{pET} - \text{PE} \quad (1)$$

Daily PE (mm) was calculated by subtracting runoff and drainage from precipitation.

$$\text{PE} = P - \text{runoff} - \text{drainage} \quad (2)$$

where runoff (mm) and drainage (mm) are horizontal water loss at the soil surface and downward water loss at the bottom of the root zone, respectively.

A census for the total irrigation water abstraction in 2010 was available at NUTSIII level (a higher spatial resolution than NUTSII) although only as an aggregated value over all crops (EUROSTAT, 2014). To compare the simulated volume of NIR with the census, we aggregated the census abstraction values to the NUTSII level and compared them to the simulated volume of NIR. In each NUTSII zone, we averaged the NIR over the six crops and calculated the volumes of NIR as the product of the mean NIR depth and the total irrigated area for all crops. Although the reported volume of irrigation water abstraction included crops other than the six simulated crops, it was a useful validation for the variability of NIR volume across NUTSII zones with different irrigation requirements.

The average NIR ($\overline{\text{NIR}}$, mm) for the whole study area for each crop in each scenario was calculated as:

$$\overline{\text{NIR}} = \frac{\sum_{\text{NUTSII}=1}^{\text{NUTSII}=n} \text{NIR}_{\text{NUTSII}} \times A_{\text{IRRI, NUTSII}}}{\sum_{\text{NUTSII}=1}^{\text{NUTSII}=n} A_{\text{IRRI, NUTSII}}} \quad (3)$$

where $A_{\text{IRRI, NUTSII}}$ indicates the irrigated area for a crop in a NUTSII zone, and n is the number of NUTSII zones growing the specific crop.

The impact of climate change on the $\overline{\text{NIR}}$, $\Delta\overline{\text{NIR}}_{\text{CC}}$ was calculated as:

$$\Delta\overline{\text{NIR}}_{\text{CC}} = \overline{\text{NIR}}_{\text{CC}} - \overline{\text{NIR}}_{\text{Baseline}} \quad (4)$$

where CC indicates the SRES climate change scenarios, that is, B1, B2, or A1B.

Crops' yields under rainfed and irrigated conditions

The irrigated area and total cropping area for the six crops for 2010 were obtained from EUROSTAT (2014). These data were

available at the NUTSIII level and aggregated to the NUTSII level. For NUTSII zones with a total cropping area of A_{TC} and an irrigated area of A_{IRRI} , the irrigation ratio (IR) was calculated as $\text{IR} = A_{\text{IRRI}}/A_{\text{TC}}$. For a NUTSII zone, if only A_{IRRI} , but not A_{TC} , was reported for 2010, the value of A_{TC} from 2007 was used. For a NUTSII zone where only A_{TC} was reported or neither A_{IRRI} nor A_{TC} was reported, IR was set to 0. As a final step, the IR at the NUTSII level was disaggregated to SU level for further crop yields simulation (Fig. 3).

We simulated the yields for the six crops in each SU for 25 years with climate data both from the baseline (1982–2006) and the three climate change scenarios (2040–2064, i.e. B1, B2, A1B). The simulations were conducted under fully irrigated (FI) and rainfed conditions, which were both limited by nitrogen availability. Following Balković *et al.* (2013) and Deryng *et al.* (2014), the crop yield with a specific IR was calculated as:

$$\text{Yield} = \frac{\text{Yield}_{\text{Rainfed}} \times (A_{\text{TC}} - A_{\text{IRRI}}) + \text{Yield}_{\text{FI}} \times A_{\text{IRRI}}}{A_{\text{TC}}} \quad (5)$$

$$= (1 - \text{IR}) \times \text{Yield}_{\text{Rainfed}} + \text{IR} \times \text{Yield}_{\text{FI}}$$

where $\text{Yield}_{\text{Rainfed}}$ is the mean rainfed yield across 25 years in a specific scenario, that is, baseline, B1, B2, and A1B, for a specific SU. Yield_{FI} indicates mean yield under full irrigation.

With a specific irrigation ratio (IR) of a SU, the climate change impact on yields was calculated as:

$$\Delta\text{Yield} = \frac{\text{Yield}_{\text{CC}} - \text{Yield}_{\text{Baseline}}}{\text{Yield}_{\text{Baseline}}} \times 100; [\%] \quad (6)$$

where CC indicates different SRES scenarios, that is, B1, B2, and A1B.

Under rainfed conditions, climate change impacts on yields were calculated as:

$$\Delta\text{Yield}_{\text{Rainfed}} = \frac{\text{Yield}_{\text{Rainfed, CC}} - \text{Yield}_{\text{Rainfed, Baseline}}}{\text{Yield}_{\text{Rainfed, Baseline}}} \times 100; [\%] \quad (7)$$

where CC indicates different climate change scenarios, that is, B1, B2, and A1B.

The difference of % ΔYield under rainfed and irrigated conditions was quantified as:

$$\Delta\Delta\text{Yield} = \Delta\text{Yield}_{\text{Rainfed}} - \Delta\text{Yield}; [\text{p.p.}] \quad (8)$$

For crops with large $\Delta\Delta\text{Yield}$, the absolute value of $\Delta\Delta\text{Yield}$ was plotted against the irrigation ratio, IR. Note the yields presented for irrigated conditions is not full irrigation, but rather the weighted averaged of full irrigation and rainfed yields, using the NUTSII, crop-specific IR (Eqn 5).

Results

Irrigation requirement under baseline and climate change conditions

The spatial patterns of net irrigation requirement (NIR) were similar for the six crops, lower in the northern regions and higher in the southern Mediterranean regions (Fig. 4). For the three winter crops, less than

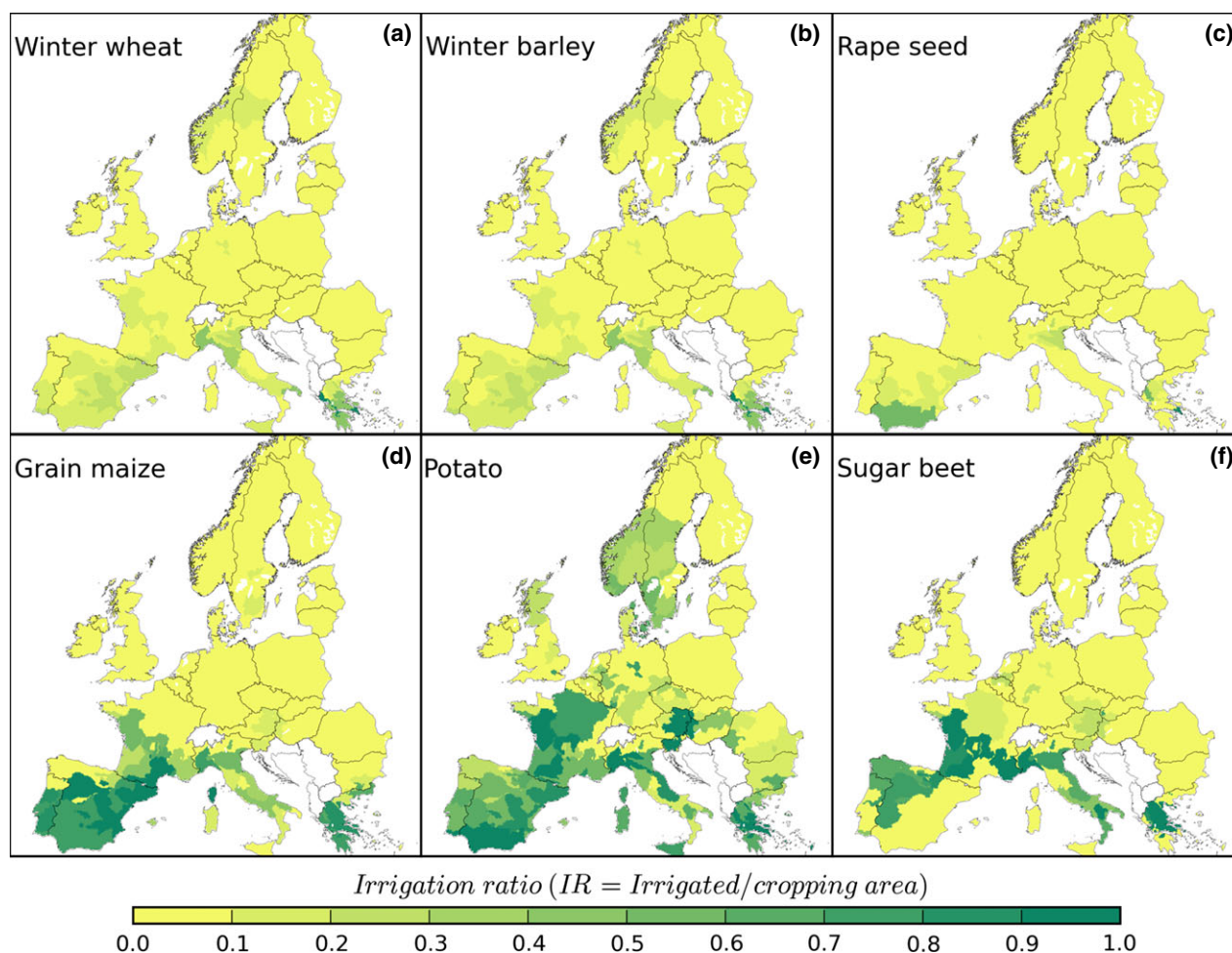


Fig. 3 Maps of irrigation ratio (IR) for six crops in year 2010 for winter wheat (a), winter barley (b), rapeseed (c), grain maize (d), potato (e), and sugar beet (f). IR indicates the irrigation ratio for a NUTSII zone.

100 mm yr⁻¹ of irrigation water was required in central and northern regions, but in the southern Mediterranean countries, the NIR reached more than 600 mm yr⁻¹ in some cases. In general, the NIR for the three spring crops was higher than for the three winter crops. In the mid-latitudes, NIR attained 400 mm yr⁻¹, while in the Mediterranean countries, NIR of maize reached as much as 836 mm yr⁻¹. The simulated volumes of NIR were consistent with the reported irrigation water abstraction in most of the NUTSII zones with regard to both the spatial pattern and value range (Fig. 5).

The spatial patterns of Δ NIR were similar across the three SRES scenarios, but with different magnitudes (Figs 6, S5 and S6). In the A1B scenarios, NIR decreased by up to 81 mm yr⁻¹ (w/CO₂) in mid-latitude to high-latitude regions (e.g. Northern Germany and Finland) for the three winter crops, but increased by up to 60 mm yr⁻¹ in most of the Mediterranean countries (e.g. Spain and Italy) (Fig. 6). The Δ NIR was small in

northern France, Southern Germany, and Austria (–10 to 10 mm yr⁻¹). For the three spring crops, increasing NIR was observed across most regions, especially when the CO₂ effect on transpiration was not considered. For grain maize, the largest increase (up to 182 mm yr⁻¹ w/o CO₂) occurred across many regions in Portugal, western France, and Italy. The increases for the other areas varied from 10 to 40 mm yr⁻¹. There was a small decrease of NIR in southern Spain for grain maize. Compared to the other two spring crops, the increase of NIR for potato was much smaller across all regions, among which the largest increase of 70 mm yr⁻¹ occurred in eastern France. Similar to grain maize, a small decrease of NIR also occurred in southern Spain. Without considering the CO₂ effect on transpiration, a very large increase of NIR occurred for sugar beet, which exceeded 100 mm yr⁻¹ in about half of the regions. Consideration of the effect of CO₂ on transpiration resulted in a trend change (from increase to decrease decreased) for NIR of sugar beet and grain

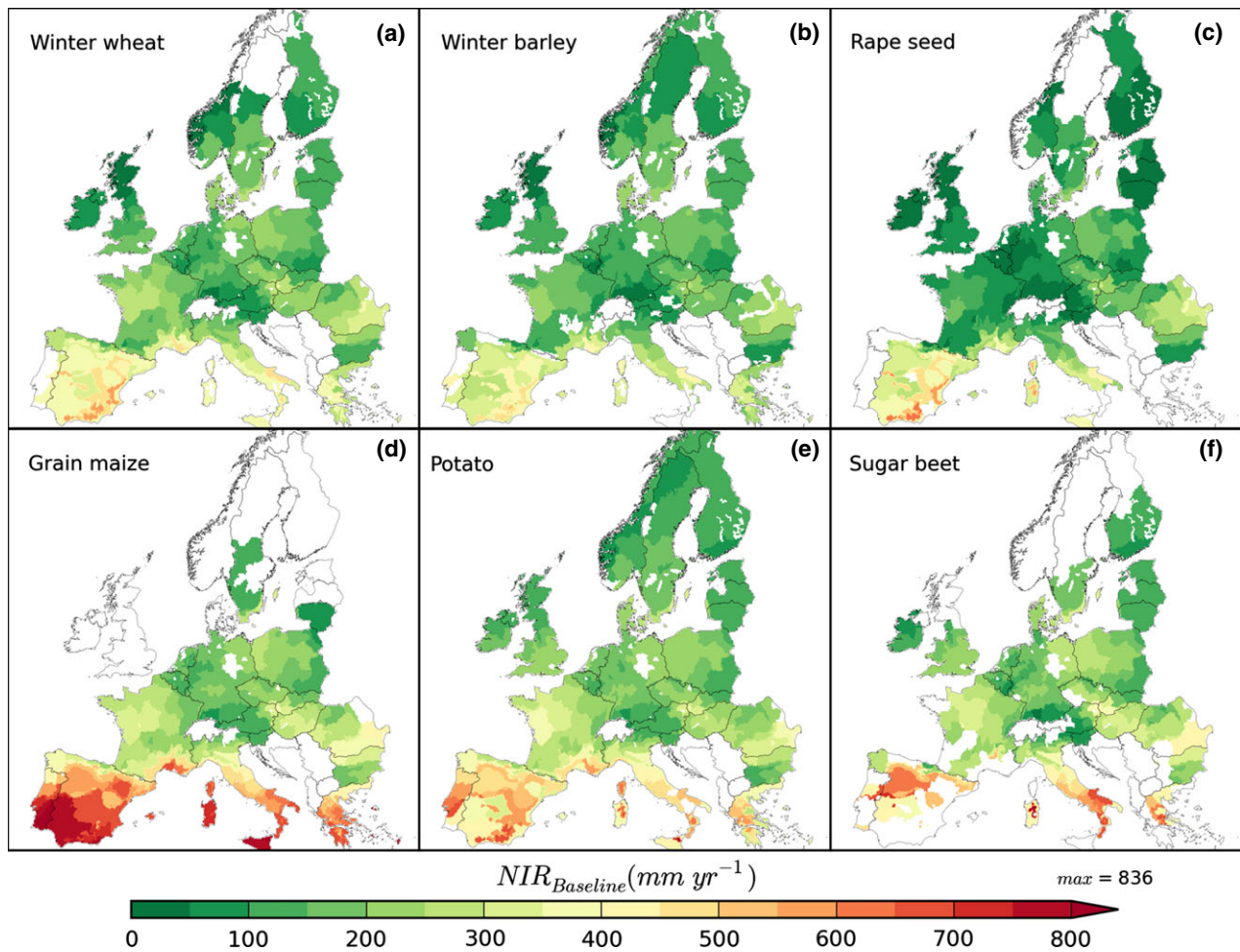


Fig. 4 The simulated net irrigation requirement (NIR, mm yr^{-1}) (mean across years) for six crops in the baseline period (1982–2006). The six crops include winter wheat (a), winter barley (b), rapeseed (c), grain maize (d), potato (e), and sugar beet (f). IR indicates the irrigation ratio for a NUTSII zone.

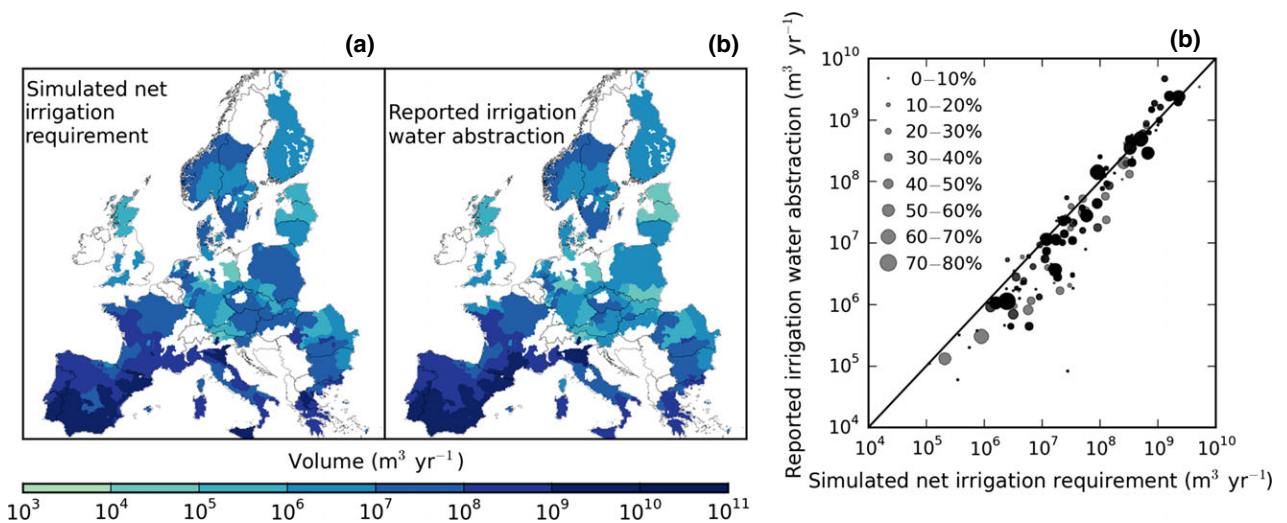


Fig. 5 Comparison of simulated net irrigation requirement (a) with the reported irrigation water abstraction (b) at NUTSII level in the baseline. The blank NUTSII zones in the map have no reported irrigation water abstraction. The dot size in the scatter plot (c) indicates the area percentage that the six crops represent, that is, the ratio of the irrigated area of the six crops to all the crops.

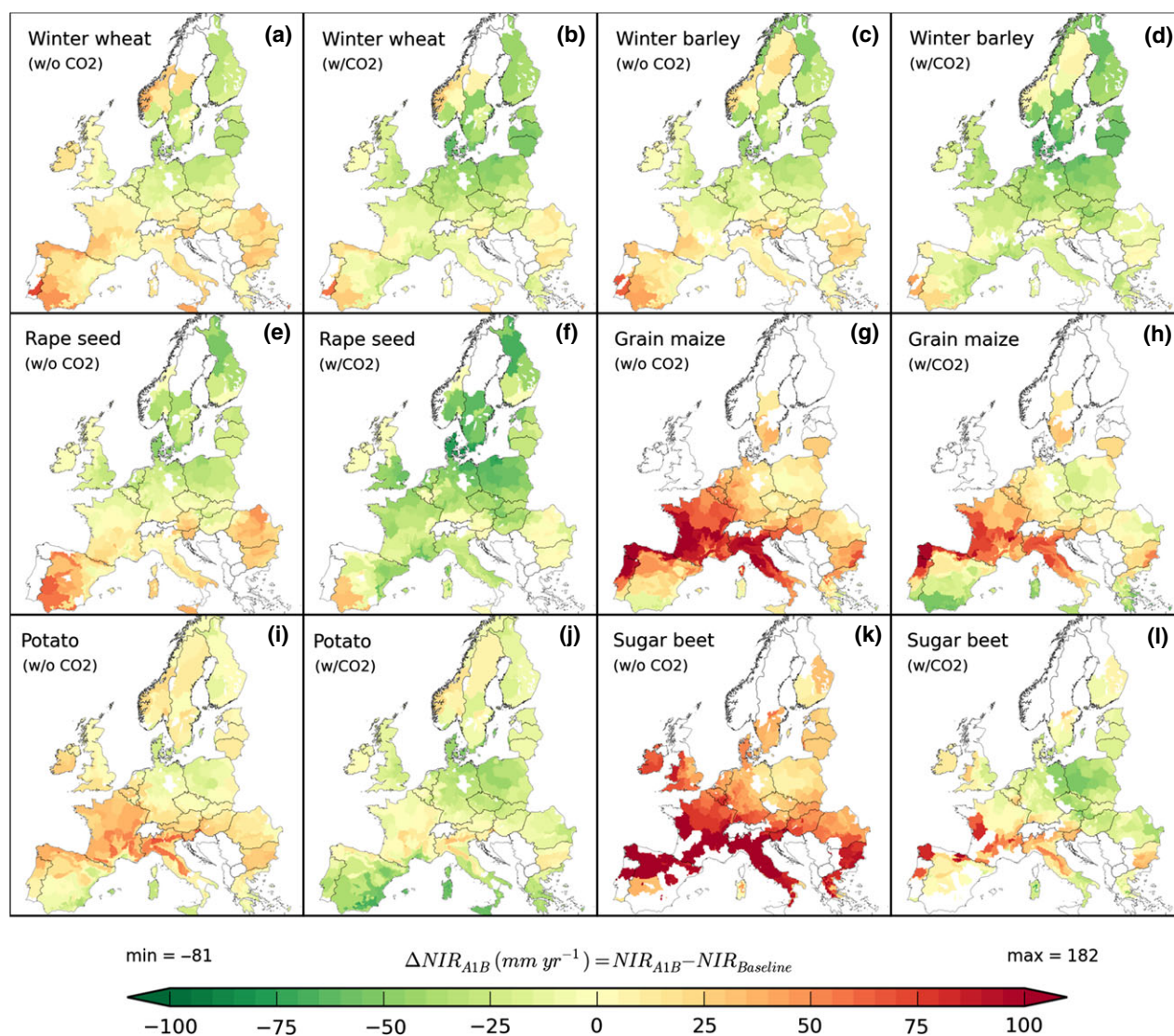


Fig. 6 Simulated impacts of climate change on the net irrigation requirement (ΔNIR) of six crops in the SRES A1B scenario across simulation units in EU25. The six crops are winter wheat (a and b), winter barley (c and d), rapeseed (e and f), grain maize (g and h), potato (i and j), and sugar beet (k and l). Simulations are performed without considering CO₂ effects on crop growth and transpiration considering (a, c, e, g, i, and k) and with CO₂ effects considered (b, d, f, h, j, and l).

maize in the middle-east of Europe. Consideration of CO₂ effects also further decreased NIR for the three winter crops in the northern regions.

Averaged across all currently irrigated croplands, without consideration of the CO₂ effect on transpiration, projected climate change resulted in slightly decreased NIR for rapeseed in the B1 scenario (4 mm yr⁻¹), but led to increases of varying magnitudes for the other five crops and rapeseed in the B2 and A1B scenarios (Fig. 7). The largest average increase of NIR (72 mm yr⁻¹) was calculated for sugar beet in the A1B scenario. The magnitudes of the increases differed across crops and climate change scenarios. Three crops had the largest increases in the B1 scenario (e.g.

winter wheat, winter barley, and potato), whereas other crops had the largest increases in the A1B scenario (e.g. rapeseed, grain maize, and sugar beet). The CO₂ effect on transpiration changed the direction of average change for winter barley and rapeseed in B2 and A1B scenarios, potato in all three scenarios, and sugar beet in B1 scenario.

Spatial-temporal variability of simulated yield under the baseline condition

Across all six crops, the model captured the spatial variability of observed yields (1984–2006), particularly when irrigation was considered (Fig. 8a–r). Over most

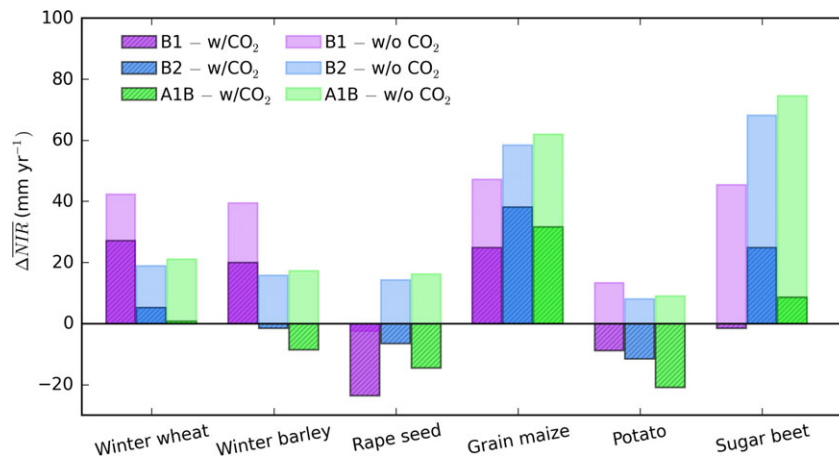


Fig. 7 Simulated average European-wide change of net irrigation requirement for six crops under three climate change scenarios in the current irrigated area. The changes were calculated using Eqn 5. The bars with hatch and color filled indicate the change of NIR with the CO₂ effects on crop growth and transpiration considered. The bars with only color filled indicate the change of NIR without the CO₂ effects on crop growth and transpiration considered.

areas of Europe, the model captured more than 40% of the temporal variation for the six crops (Fig. S9). For the three winter crops (i.e. winter wheat, winter barley, and rapeseed), the model simulated similar yields with either irrigated or rainfed conditions (Fig. 8b, c, e, f, h, i). However, for the three spring crops under rainfed conditions, the model notably underestimated the yields in many Mediterranean regions where the irrigation ratio was high, for example, Spain and Greece (Figs 8k, n, q and S10) and slightly overestimated yields when actual irrigation was considered across these regions (Fig. S10).

For the aggregated yields, under rainfed and irrigated conditions, the model captured the interannual yield variability for all crops across the 25 years in the baseline period, for example, low yield in 2003 (Fig. 8s–x). The correlation coefficients varied from 0.22 (winter barley) to 0.74 (grain maize), and irrigation did not improve the correlation ($\Delta\rho < 0.03$). The model overestimated the magnitude of interannual variability for the three spring crops. Irrigation only slightly changed the interannual variability ($\Delta\rho < 0.03$) and yield levels for winter crops by 0.16 t ha^{-1} (4%), 0.09 t ha^{-1} (3%), and 0.03 t ha^{-1} (1%) for winter wheat, winter barley, and rapeseed, respectively. It also slightly changed the interannual variability of the aggregated yields for the three spring crops, but notably changed the yield levels by 1.5 t ha^{-1} (42%), 1.4 t ha^{-1} (27%), and 2.5 t ha^{-1} (25%) for grain maize, potato, and sugar beet, respectively. The model underestimated the yields of barley by 1.0 t ha^{-1} . For the three spring crops, the observed yields were between the simulated yields under irrigation (marked as simulated in the figure) and

rainfed conditions. On average, the model underestimated yields if rainfed conditions were assumed.

The results of the spatially explicit correlation and root mean square error (RMSE) analysis between observed and simulated yields (Figs S9 and S10) showed that accounting for irrigation improved the correlation for the three spring crops in the Mediterranean countries, but did not improve the RMSE. For the three winter crops, the influence of irrigation is very small for most regions with respect to both the correlation coefficients and RMSE.

Yield changes under projected climate change conditions

Without CO₂ effects for the three winter crops in the A1B scenario, climate change would have considerable negative impacts (up to 49%) on yields in most southern Mediterranean countries, but small positive impacts in the mid-latitude (42°N–55°N) and Nordic countries (Fig. 9a, c, e). CO₂ effects altered the direction of the impacts from negative to positive (up to 36%) for regions from northern France toward Nordic countries (Fig. 9a, c, e). By considering CO₂ effects across Europe, the total area with increased yields was larger than the area with yield declines. For the three winter crops, the greatest yield increase occurred in the UK, northern Sweden, Germany, Estonia, Latvia, and Lithuania.

Without consideration of CO₂ effects, negative impacts of climate change were much less pronounced for the spring crops than for the winter crops in Mediterranean countries (Fig. 9g–i). CO₂ effects moderated the negative impacts of climate change on the yield for grain maize in the middle and northern regions (Fig. 9h), except for some Mediterranean

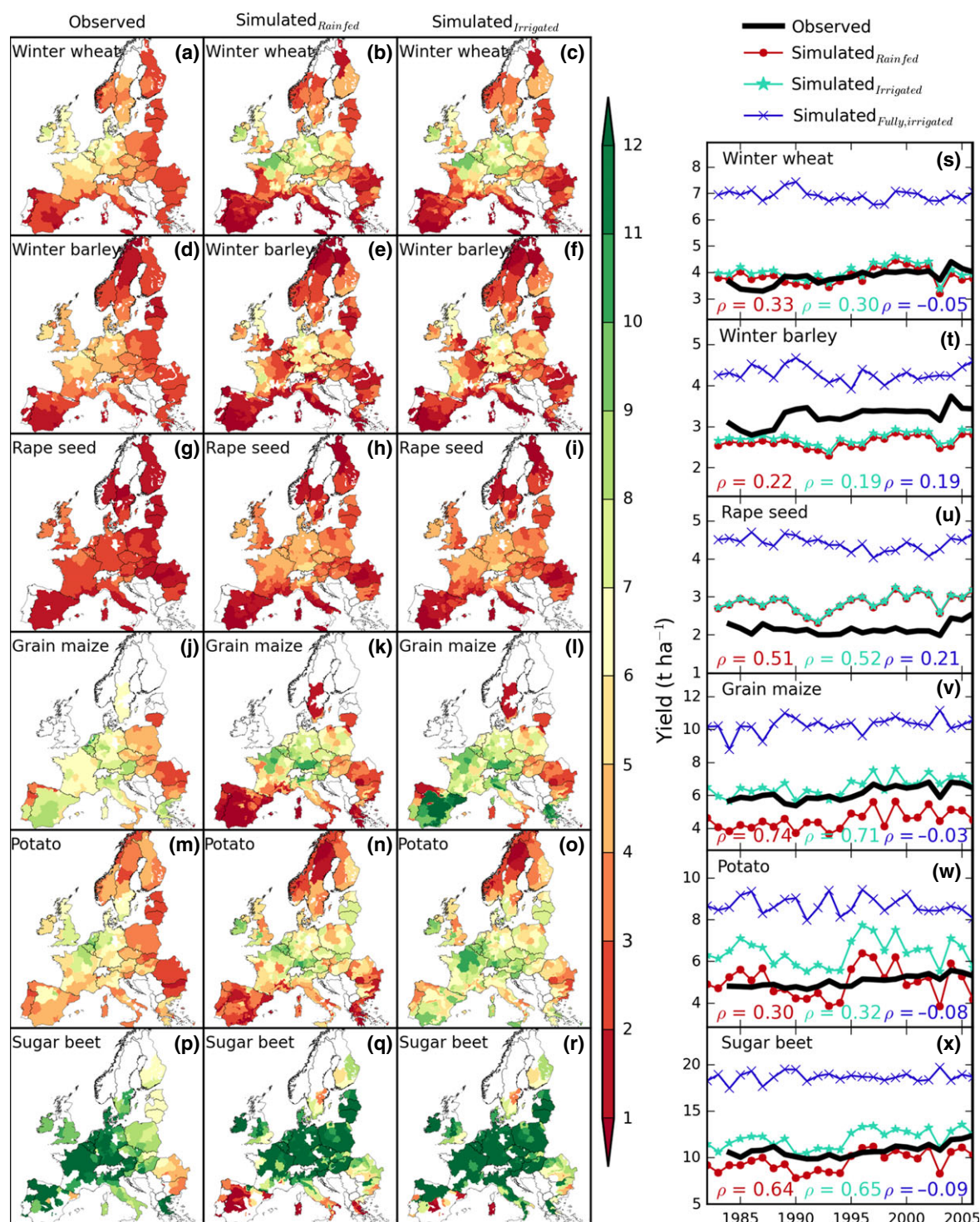


Fig. 8 Spatial-temporal variability of observed (a, d, g, j, m, and p), simulated-rainfed (b, e, h, k, n, and q), simulated-irrigated (actual irrigation was considered, c, f, i, l, o, and r), and simulated-fully-irrigated yields (only time series) in the baseline period (1982–2006) for the six crops. The six crops are winter wheat (a, b, and c), winter barley (d, e, and f), rapeseed (g, h, and i), grain maize (j, k, and l), potato (m, n, and o), and sugar beet (p, q, and r). Note under all of the three conditions, the simulated yield was also limited by availability of nitrogen. The maps display mean yields across the baseline years (1982–2006). The time series plots display the mean of yields over the whole study area (s, t, u, v, w, and x). The observed yields were obtained from EUROSTAT (2014). The ρ values at the bottom of the time series plots are the Spearman's correlation coefficients between simulated (from left to right, simulated-rainfed, simulated-irrigated, and simulated-fully-irrigated yields) and observed yields. [This figure is colour in the online version]

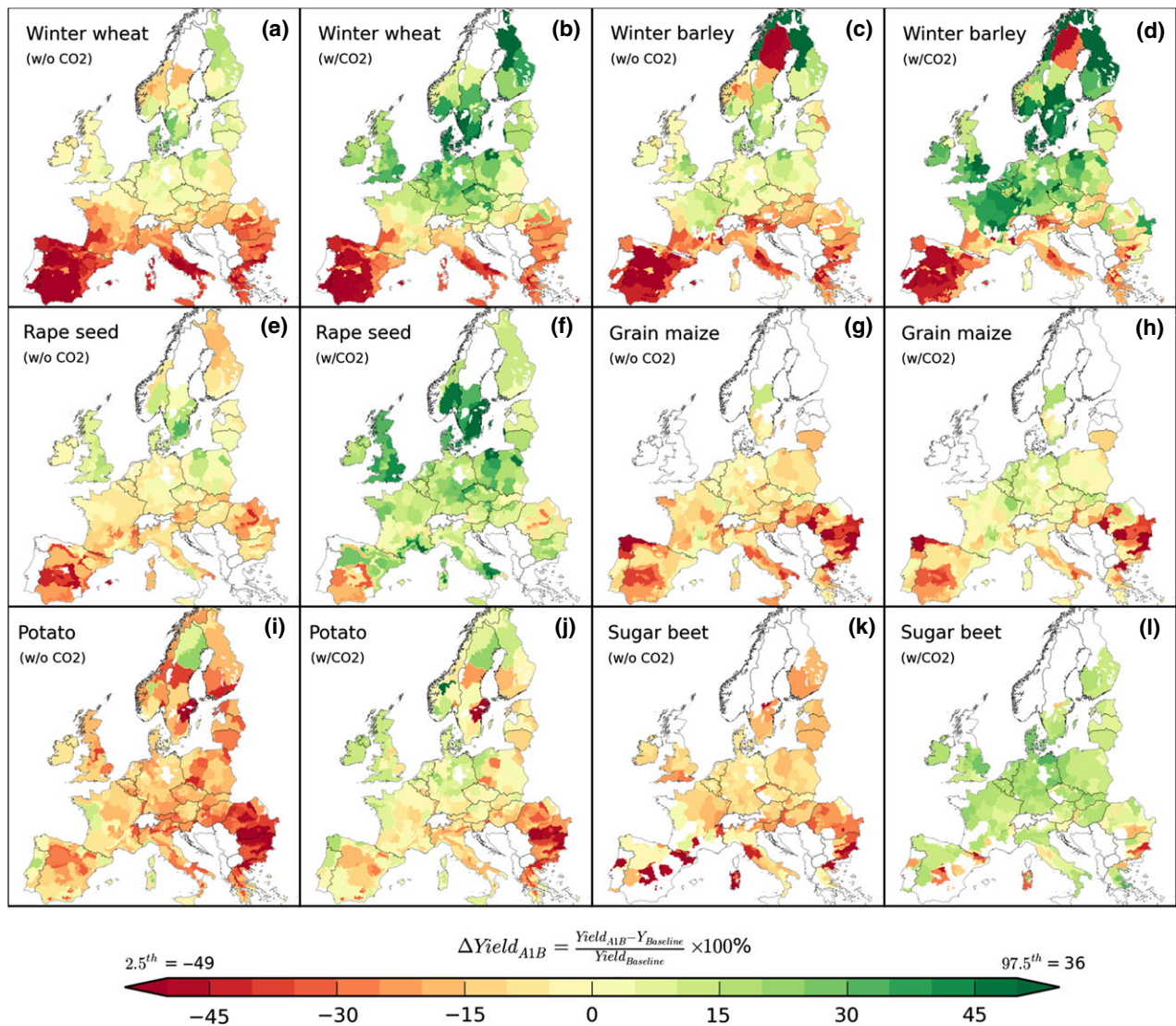


Fig. 9 Simulated impacts of climate change on the yields of six crops in the SRES A1B scenario across simulation units in EU25. Irrigation was considered in the simulations. The six crops are winter wheat (a and b), winter barley (c and d), rapeseed (e and f), grain maize (g and h), potato (i and j), and sugar beet (k and l). Simulations are performed without considering CO₂ effects on crop growth and transpiration considering (a, c, e, g, i, and k) and with CO₂ effects considered (b, d, f, h, j, and l).

countries, for example, Spain and Bulgaria. CO₂ effects altered impacts from negative to positive for sugar beet for most regions in Europe (Fig. 9l). The spatial patterns and signs of yield change were consistent across the three projected climate change scenarios, but the magnitudes of the changes differed (Figs 9, S7 and S8).

Comparing yield changes between rainfed and irrigated conditions

The inclusion of actual irrigation in the simulation of relative yield change had essentially no impact for the three winter crops (Fig. 10a–c), but had positive effects for the three spring crops (Fig. 10d–f). The magnitudes

of the bias ($\Delta\Delta\text{Yield}$) were consistently in proportion to the irrigation ratio (Fig. 11).

Discussion

Climate change, especially higher temperatures, shifting patterns of rainfall, and enrichment of atmospheric CO₂ concentration, will inevitably affect future crop growth environments and production (Goubanova & Li, 2007). Irrigation, as a commonly used management practice in many arid and semi-arid regions of Europe, plays an important role in shaping crops' response to climate change. Using a process-based crop model, SIMPLACE <LINTUL5, DRUNIR, HEAT>, we assessed

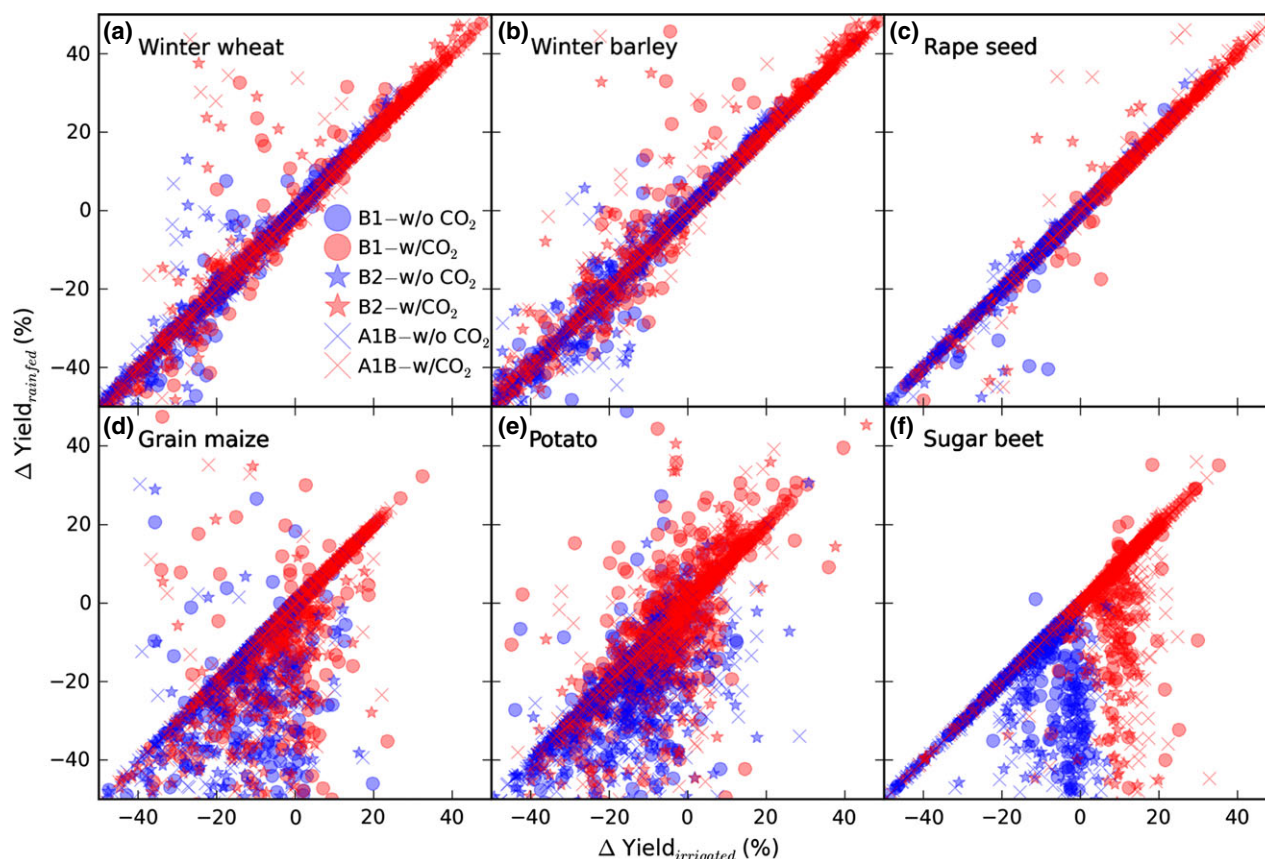


Fig. 10 Simulated differences in yield change between rainfed (y axis) and irrigated (x axis) conditions for six crops in three projected climate change scenarios, that is, B1 (dot), B2 (star), and A1B (multiplication sign), with (blue) and without (red) considering the CO₂ effects on crop growth and transpiration. The six crops include winter wheat (a), winter barley (b), rapeseed (c), grain maize (d), potato (e), and sugar beet (f).

the impacts of climate change on the irrigation requirement of six arable crops across Europe and the implication of considering irrigated vs. rainfed conditions in climate change impact assessments for crop yield, which has not been done before. We also explicitly quantified the uncertainty in crop yield and water use estimates caused by considering or not CO₂ effects on crop growth and transpiration. The results provide useful information about the importance of considering irrigation and CO₂ effects in climate change assessment studies. The proposed methodology should be applicable to other regions and crops.

Model performance and uncertainty analysis

The comparison of reported irrigation water abstraction and simulated NIR across NUTSII zones showed that the model agreed well with respect to both the magnitude and spatial variability (Fig. 5). Simulated NIR in Spain was in the same range as that calculated by Rodríguez Díaz *et al.* (2007). The spatial variability and

magnitude of the current NIR were also close to results simulated by the EPIC model (Wriedt *et al.*, 2009). Although good agreements were simulated, several caveats for the interpretation of the simulated and observed values should be pointed out. First, the reported total water abstraction was a product of irrigation requirement of many crops, irrigation efficiencies, and irrigation levels, while we simulated net irrigation requirement of six crops. On the one hand, in many regions of Europe, deficit irrigation is a common practice (Di Paolo & Rinaldi, 2008) which decreases water abstraction. This could make the NIR higher than the reported water abstraction. On the other hand, irrigation efficiencies vary across regions ranging from 0.41 (low efficiency) to 0.70 (high efficiency), and lower irrigation efficiency leads to higher water abstraction (Wriedt *et al.*, 2008). The under- or overestimation due to these reasons may cancel out each other and result in similar values when comparing the simulated NIR and reported irrigation water abstraction (Fig. 5). Moreover, we only simulated NIR for six crops and

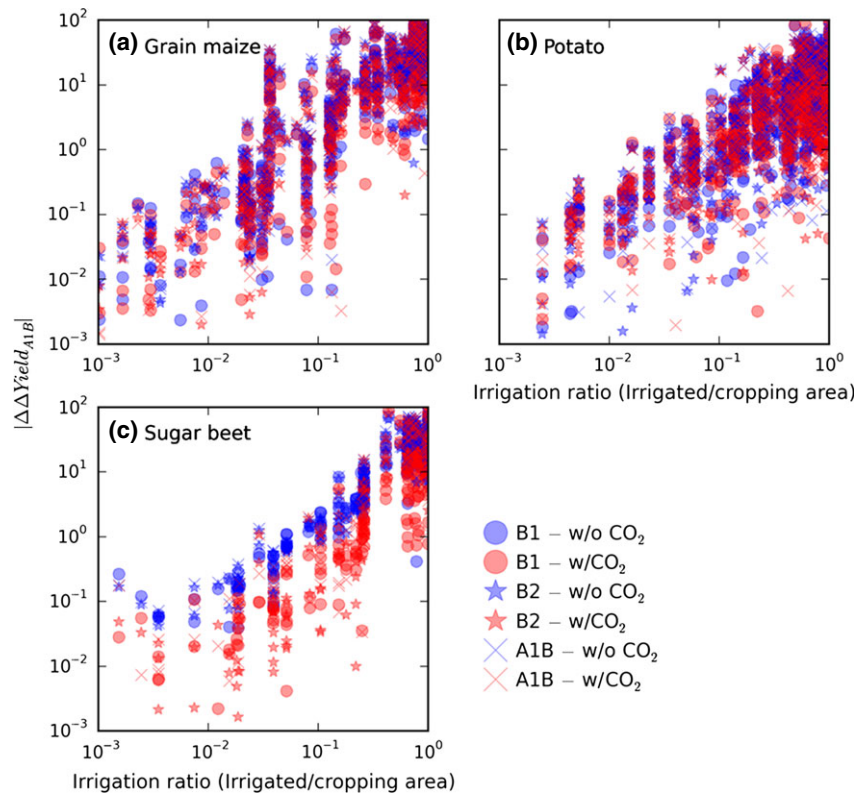


Fig. 11 Relationships between the irrigation ratio and corresponding differences of yield change between rainfed and irrigated conditions for three crops in three projected climate change scenarios, that is, B1 (dot), B2 (star), and A1B (multiplication sign), with (blue) and without (red) considering the CO₂ effects on crop growth and transpiration. The three crops including grain maize (a), potato (b), and sugar beet (c) were chosen according to the differences in Fig. 10. The calculation of $\Delta\Delta\text{Yield}$ refers to Eqn 8.

representations of the six crops compared with all crops vary from 10 to 80% across NUTSII zones (Fig. 5c). The comparison may be particularly biased in the low representative zones. Finally, there are uncertainties in the reported irrigation abstraction numbers, which are based on estimates rather than real measurements (Wriedt *et al.*, 2008). In their estimation, different methods are used when measured data are not or only partially available (Nagy *et al.*, 2007). Further, illegal abstractions are frequently reported in many regions (Flörke *et al.*, 2010), with an estimated 3.6 million m³ water or more being illegally abstracted annually in Spain (WWF/Adena, 2006). All factors contribute to uncertainties in the reported irrigation abstraction volumes. Nevertheless, the simulated NIR was closely correlated with the observed value and the magnitudes were also in the same range, justifying the use of SIMPLACE <LINTUL5, DRUNIR, HEAT> for simulating the NIR for the six crops across Europe.

The simulated yields can explain more than 40% of the temporal variability across the majority areas of the six crops (Fig. S9), although under- or overestimated yields occurred in many regions, especially for grain maize, potato, and sugar beet (Figs 8 and S10).

Consideration of irrigation in the simulations improved the correlation for the three spring crops in the Mediterranean countries, but not for the winter crops nor for the root mean square error (RMSE). This was due to the fact that irrigation ratio was very low for the winter crops (Fig. 3). Additionally, we used one year irrigation data for the 25-year simulation. The deficit irrigation and temporal change of irrigation were also not considered due to the lack of information. Therefore, it is expected that the correlation and RMSE cannot be significantly improved.

In this study, we only calibrated the thermal requirements from sowing to flowering and from flowering to maturity. The values for the other parameters were obtained from Boon-Prins *et al.* (1993). This calibration strategy ensured that the model could reproduce the observed flowering and maturity dates and took account of the cultivar difference across the environmental zones. A similar calibration strategy was also adopted in Therond *et al.* (2011), Balkovič *et al.* (2013), and Angulo *et al.* (2013). Therond *et al.* (2011) calibrated LINTUL in the APES modeling framework based on the temperature sum of the average sowing and harvest dates in 12 EU regions and notably improved the

model's performance in simulating yields for three crops. In addition to the similar strategy of phenology calibration, Angulo *et al.* (2013) used the drought tolerance parameter of LINTUL to calibrate the model for the severe drought stress conditions in the Mediterranean countries. As the drought tolerance parameter did not vary across regions, the model overestimated the yields in regions where drought stress was high but with no irrigation. The use of the spatially and crop-explicit irrigation ratio effectively avoided these drawbacks. Similarly, Balković *et al.* (2013) found that spatially explicit management and cultivar characteristics could improve the simulation of yields for four crops in Europe using EPIC.

Several other sources of uncertainty in the simulations need to be pointed out here. First, irrigation was only used to avoid drought stress in the simulations. In reality, it also has a cooling effect for the ambient and canopy temperature (Jackson *et al.*, 1977; Throssell *et al.*, 1987), thus reducing heat stress during the sensitive development stage after flowering. Ignoring the cooling effects of irrigation could overestimate heat stress, simulating more negative climate change effects (Eyshi Rezaei *et al.*, 2015; Lobell *et al.*, 2008). Secondly, irrigation could increase the nitrate leaching in sandy soils and enhance stress from N availability (Eitzinger *et al.*, 2013). However, our model did not simulate the interaction between irrigation and N leaching. This could result in an overestimation of yields on sandy soils when irrigation is applied. Also, we only used data on irrigated and cropped areas for 2010 to develop the data of irrigation ratios for different crops. The irrigation ratio and the spatial extent do not fully represent historical irrigation conditions. This may have led to uncertainties in simulating the yields and NIR in the baseline (Wisser *et al.*, 2008). Finally, we use only one model for the NIR and yield simulation, while the uncertainty and discrepancies are normally large across models (Asseng *et al.*, 2013; Bassu *et al.*, 2014; Li *et al.*, 2015). A relatively large number of crop models are needed to achieve a level of confidence consistent with yield variation expected with observations, for example, 4–14 models for different warming scenarios (Asseng *et al.*, 2013), which was not in the scope of this study.

Change of irrigation requirement under projected climate change conditions

The changes in net irrigation requirement (ΔNIR) due to projected climate change differed notably across the six crops (Figs 6, 7, S5, and S6), especially between the spring and winter crops, which was due to the timing and duration of crop-specific processes. Warming

temperatures shortened the growing season length (GSL) for the six crops, but much more so for grain maize and sugar beet than the other crops (Fig. S10). Shortened GSL resulted in less time to consume water by evapotranspiration during the growing season (Fig. 12). However, in cases without consideration of the CO_2 effect on transpiration, higher temperatures and increased solar radiation could lead to greater cumulative evapotranspiration and result in a large increase in NIR in some regions and scenarios despite the shorter growing season (Fig. 12a, b, f, w/o CO_2). The winter crops in the northern regions grow during a relatively cool period of the year, during which the effects of decreased solar radiation on potential ET overcame the effects from increased temperature, leading to decreased potential evapotranspiration. Additionally, slightly higher rainfall in this region also resulted in decreased NIR under projected future climate change (Fig. 2g–i). However, the decrease in effective rainfall significantly surpassed the decrease in potential evapotranspiration during the growing season of spring crops and resulted in increasing NIR in the same regions (Fig. 12d–f, w/o CO_2). The differences between winter and spring crops imply that crop-specific simulation (e.g. phenology) is essential in estimating the NIR at regional scale (Wriedt *et al.*, 2009).

The differences across crops point to the opportunity of using crops that demand less water in regions with limited water resources (Weatherhead & Knox, 2000; Iglesias *et al.*, 2007). In the southern dry regions, potential evapotranspiration increased due to increased temperature and radiation and less effective rainfall, which resulted in a significant increase of NIR for all six crops. In line with this study, Trnka *et al.* (2011) predicted a significant increase of water deficit under projected climate change conditions (A2) in the environmental zones of LUS, MDM, MDS, and MDN for both winter and spring crops and inferred that the additional water requirement may not be satisfied by the availability of water resources. Rodríguez Díaz *et al.* (2007) predicted a 12% increase of NIR in the Guadalquivir river basin in Spain under the B2 scenario in 2050. For a large increase of NIR for sugar beet, in line with this study, Jones *et al.* (2003) found that drought stress could double in northeastern France and Belgium and irrigation requirements will increase by up to 18% in central Europe in the period 2021–2050 compared to the baseline. Maracchi *et al.* (2005) recommended earlier sowing dates or cultivar for spring crops to take the advantage of warming temperature, avoid heat and drought stress in the summer, and reduce the NIR in southern Europe. In addition to adaptive management practices, policies such as irrigation water pricing could be used to control irrigation water use, sustain agricultural production,

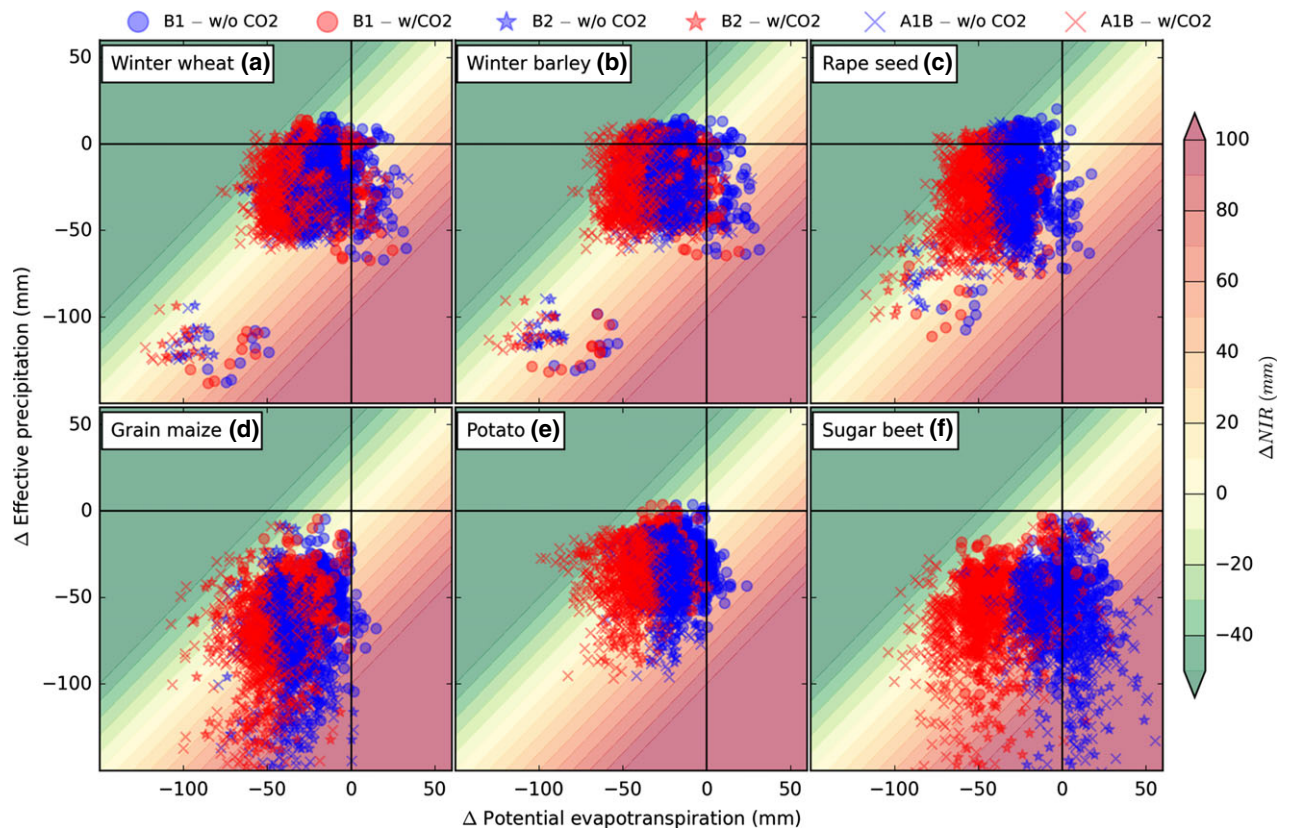


Fig. 12 Relationships between changes of potential evapotranspiration, effective precipitation, and net irrigation requirement (NIR) for six crops and three projected climate change scenarios, that is, B1 (dot), B2 (star), and A1B (multiplication sign), with (blue) and without (red) considering the CO₂ effects on crop growth and transpiration. The six crops include winter wheat (a), winter barley (b), rapeseed (c), grain maize (d), potato (e), and sugar beet (f). IR indicates the irrigation ratio for a NUTSII zone.

and moderate the conflicts for water use among various socio-economic aspects (Manos *et al.*, 2006; Bournaris & Manos, 2012).

We found that CO₂ effects may alter the direction of NIR change for various regions for grain maize and sugar beet. In Europe, few studies have considered the CO₂ effect on transpiration, which were found as a major source of uncertainty here and in global impact assessments, for example, Rosenzweig *et al.* (2013) and Elliott *et al.* (2014). As we derived parameters for CO₂ effect on transpiration from FACE experiments for different crops, our study captured more crop-specific responses and interactions under the various environmental conditions comparing to Angulo *et al.* (2013) which used the same response for all crops.

Impact of climate change on crops' yields and the role of irrigation

Warmer temperatures shorten the crop growth duration (Fig. S11) and increase crops' vulnerability to heat stress in southern regions, but increase RUE in the

cold seasons in northern regions. Therefore, the crops growing in cold seasons in the high-latitude regions may benefit from the warming temperatures. This is supported by the significant increase of yields for winter crops but not for the spring crops in the cold northern regions under all three projected climate change scenarios (Fig. 9). As the growing period of spring crops falls into the warm season and the solar radiation was projected to increase in the south but to decrease in northern regions with different magnitudes across three scenarios (Fig. 2d–f), the potential yield may increase to a similar extent when the effects from increased solar radiation surpass the effects due to shortening of the growing season. However, precipitation notably decreased in the southern regions which further enhanced the irrigation requirement in these regions where there is already intensive irrigation. Again, these effects were partly compensated for by the effects of increased CO₂ concentration on the crop growth, explaining the considerable spatial variability of yield change across regions, crops and scenarios.

Similar to NIR, the uncertainty arising from CO₂ effects on yields was large enough to alter the sign of relative yield changes across many regions in Europe (Figs 9, S7 and S8). Similar to the present study, Torriani *et al.* (2007) found that climate change may negatively affect the mean maize yields, but increase the mean yield of winter wheat with elevated CO₂ concentration. Iqbal *et al.* (2011) found that projected climate change decreased maize yield regardless of whether CO₂ fertilization effects were considered in Faisalabad, Pakistan, using the CERES-Maize model. Using climate data from an ensemble of five GCM models, Alexandrov *et al.* (2002) found that climate change would result in a yield decrease for winter wheat in Austria when CO₂ fertilization effects were not considered, but an increase otherwise.

A few studies have investigated the potential effects of irrigation in climate change impact studies, although most have not considered irrigation in regional crop model applications. For instance, Van Der Velde *et al.* (2009) ignored irrigation in assessing the water and nutrient use efficiency for rapeseed in Europe. The results from the present study indicated that assuming rainfed conditions can introduce bias into yield simulations (Fig. 8) and yield changes (Fig. 10), especially for highly irrigated crops and regions. This bias was negative and proportional to the irrigation ratio (Fig. 11).

The results from this study show that, for the three spring crops, yield changes under irrigated conditions were more positive than under rainfed conditions (Fig. 10). This is due to the fact that the irrigation ratio and irrigation requirements of the spring crops are generally high. Under the projected climate change conditions, warming temperatures will increase potential evapotranspiration during the growing season (Fig. 12). Hence, rainfed crops may experience more severe drought stresses under climate change. However, when irrigation is considered, drought stress is avoided with a result that more positive impacts of climate change were simulated. Consistent with the present study, in northeastern Austria, Alexandrov *et al.* (2002) found that yield changes due to climate change for irrigated soybean simulations were higher than those for rainfed simulations, especially when the CO₂ fertilization effects were excluded.

To conclude, to the best of our knowledge, this study provides the first assessment of impacts of climate change on the irrigation requirement for six crops in Europe. It also assesses the implication of considering present irrigation conditions and uncertainty from CO₂ effects in assessing the impacts of climate change on crop yields. The applied crop model, SIMPLACE <LINTUL5, DRUNIR, HEAT>, captured the observed spatial-temporal variability of yields for the crops con-

sidered as well as the NIR across NUTSII regions when compared to census of irrigation water abstraction data. Including spatially crop-specific irrigation ratios improved simulations in the Mediterranean countries for highly irrigated spring crops. Ignoring irrigation in the simulations led to a considerable bias in projected climate change impacts on crops' yields, and the magnitude of this bias was proportional to the irrigation ratio of the crop in a specific simulation unit. From this, we conclude that in climate change impacts studies, irrigation cannot be ignored for highly irrigated crops and regions. Under climate change, irrigation requirements of winter crops decreased in the central to northern regions of Europe but increased in the southern Mediterranean regions. However, for grain maize and sugar beet, the irrigation requirement increased considerably for the majority of areas across southern Europe. CO₂ effects are major sources of uncertainty for climate change impacts simulation of water demand and yield. These results suggest that water-saving strategies will become even more important in Mediterranean regions, which already experience water scarcity, under climate change.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. The phenological parameters for simulated six crops (winter wheat, winter barley, rape seed, grain maize, potato and sugar beet).

Table S2. The used parameters for simulating effects of CO₂ enrichment on crop transpiration.

Table S3. Interpolated correction factor of CO₂ effects on crop transpiration for six crops under different projected climate change scenarios, B1, B2 and A1B.

Figure S1. Maps of soil physical properties across European Union 25 (EU25).

Figure S2. Responses of crop radiation use efficiency (RUE) to the CO₂ concentration for C3 (red lines) and C4 (blue lines) crops in SIMPLACE <LINTUL5, DRUNIR, HEAT> for three SRES climate change scenarios, i.e. B1, B2, A1B. The CO₂ concentrations are set as 487, 478, 532, respectively, for B1, B2, and A1B.

Figure S3. Phenology observations (sowing, flowering and harvest, column 1 – 3) and calculated thermal requirements (emergence to flowering and flowering to maturity, column 4 – 5).

Figure S4. Maps of mean of historical nitrogen input rates for the simulated six crops across simulation units in EU25.

Figure S5. Simulated impacts of climate change on the net irrigation requirement (Δ NIR) of six crops in the SRES B1 scenario across simulation units in EU25.

Figure S6. Simulated impacts of climate change on the net irrigation requirement (Δ NIR) of six crops in the SRES B2 scenario across simulation units in EU25.

Figure S7. Simulated impacts of climate change on the yields of six crops in the SRES B1 scenario across simulation units in EU25. Irrigation was considered in the simulations.

Figure S8. Simulated impacts of climate change on the yields of six crops in the SRES B2 scenario across simulation units in EU25. Irrigation was considered in the simulations.

Figure S9. Temporal correlations (Spearman's Rho) between simulated and observed yields for six crops from 1984 to 2006 under rainfed (a, c, e, g, i, k) and irrigated conditions (b, d, e, f, h, j, l).

Figure S10. Root mean square errors between simulated and observed yields for six crops from 1984 to 2006 under rainfed (a, c, e, g, i, k) and irrigated conditions (b, d, e, f, h, j, l).

Figure S11. Changes of growing season length (Δ GSL) under the SRES A1B scenario for six crops across simulation units in EU25.